

ENTERPRISE PRODUCT SPECIFICATION

MobiusEMS: Unified Workforce & Commercial Operations Platform

Connecting daily project delivery, geofenced attendance, capability intelligence, objective performance, and territory sales into one role-aware workspace.

Product Version: v2.0-PROD | **Deployment:** <https://employee.whalexy.com> | **Company:** Mobius Bloom Venture Pvt Ltd

Target Audience: Prospective Enterprise Clients, CXOs, HR Directors, Operations Leads, Commercial Sales Leadership

1. Executive Summary: Breaking the "5-Tool Trap"

In most growing companies, everyday operations are fractured across single-purpose tools that do not talk to each other:

- **Hardware Attendance Clocks:** Expensive to maintain, prone to hardware failures, and useless for field sales or distributed branch offices.
- **Disconnected Project Boards:** Good for logging tasks, but completely isolated from employee records, payroll, and appraisals.
- **Spreadsheet-Driven Appraisals:** Conducted once a year long after the work was done, suffering from recency bias, subjective favoritism, and missing work proof.
- **Standalone CRM Tools:** Expensive commercial software that knows nothing about employee capacity, leave, or actual project workload.
- **Scattered Personnel Documents:** Resumes buried in email threads, policies in shared folders, and no audit trail.

MobiusEMS replaces this tool sprawl. It provides a single, role-aware operational environment where an engineer's completed task automatically feeds their weekly contribution score, a field rep's closed deal updates their territory quota and revenue ledger, and a manager's performance review is backed by tamper-proof historical evidence.

2. Our Core Engineering Principles

Principle 1: Deterministic Math Over Black-Box AI

All performance evaluations, KPI scorecards, attendance rules, sales capacity metrics, and contribution percentiles are computed with pure mathematical formulas. AI functions strictly as an advisory drafting assistant. **The platform enforces an explicit architectural policy: AI is strictly barred from making or modifying promotion, compensation, disciplinary, or termination decisions.**

Principle 2: Respectful Accountability Over Invasive Spyware

MobiusEMS rejects employee surveillance software. There are no background keyloggers, secret screen captures, or 24/7 GPS stalking. Accountability is measured through physical presence at check-in/out via browser geofencing (300m radius) and verifiable deliverables on project tasks.

Principle 3: Built-In Data Sovereignty & Local Zero-Retention Voice

Speech-to-task recognition runs locally on your host server via faster-whisper. Spoken voice commands are converted into draft tasks on your machine, and **the ephemeral audio file is shredded immediately in code.** Voice recordings never leave your servers, are never stored on disk, and incur zero per-minute external API fees.

3. Platform Capabilities at a Glance

Module	Core Capability Delivered	Primary Users
Workforce & Org Setup	Departments, teams, designations, Employee 360 profiles, career history, and resumable onboarding.	HR, Managers, Employees
Smart Attendance	Mobile browser geofence (300m radius), 11:30 AM cutoff, 4h half-day, and leave management.	All Employees, HR Ops
Project & Task Delivery	Kanban state machine, blocker escalation, cycle time, rework metrics, and audit history.	Project Teams, Managers
Local Voice-to-Task	Zero-retention speech capture (11 Indian languages + English), Hinglish NLP, and editable draft cards.	Field Staff, Engineers
Skills & Capability Matrix	Verified vs self-reported skills, departmental heatmaps, role benchmarks, and timed assessments.	HR, Practice Leads
Objective Performance	Cascading goals, measurable KPIs, 360 reviews, deterministic math scoring, and frozen snapshots.	Leadership, Managers
Contribution Intelligence	Weekly deliverable updates, velocity metrics, contribution percentiles, and evidence coverage checks.	Team Leads, People Ops
Commercial Sales & GIS	7-tier GIS hierarchy (Global to Pincode), territory quotas, lead SLA tracking, pipelines, and revenue ledger.	Sales Reps, Sales Heads
Talent Acquisition	Job description library, candidate hiring pipelines, and automated PDF resume screening with skill matching.	Recruiters, Hiring Leads
People Growth & 1-on-1s	Structured 1-on-1 meeting workspace, internal training catalog, peer recognitions, and private to-dos.	Managers, Employees
Governance & Security	Fail-closed multi-tenancy, HttpOnly rotating sessions, authenticated private files, and audit logs.	IT, Security, Compliance

4. Module Deep-Dive: People & Daily Execution

4.1 Unified Workforce & Organizational Structure

- **Dynamic Org Hierarchy:** Model Departments, Teams, and Designations with custom reporting lines and approval paths. Modify structures dynamically without database migrations.
- **Employee 360 Longitudinal Profiles:** Single pane of glass connecting personal details, active tasks, verified skills, appraisal history, leave balances, 1-on-1 logs, and career milestones.
- **Frictionless Onboarding:** Provision via secure invitation links, temporary credentials, mandatory password rotation upon first login, and a resumable onboarding wizard.
- **Role-Based Access Control:** 5 system roles (SUPER_ADMIN, HR_ADMIN, DEPARTMENT_HEAD, MANAGER, EMPLOYEE) enforcing strict separation of duties.

4.2 Smart Geofenced Attendance & Leave Management

- **Zero-Hardware Mobile Browser Check-In:** Validates location from employee mobile/laptop browsers using native HTML5 geolocation without mandatory native app downloads.
- **Haversine Geofencing (300m Radius):** Enforces check-in within a 300-meter radius of configured office coordinates. GPS accuracy fuzzier than 200m is rejected to block location spoofing.
- **Predictable Business Rules:** Check-ins after 11:30 AM IST automatically mark as LATE. Total elapsed shift under 4 hours (240 mins) upon checkout demotes the day to HALF_DAY.
- **Clean 5-State Taxonomy:** Classifies daily records into PRESENT, LATE, HALF_DAY, ON_LEAVE, or ABSENT. Includes regularization requests for field transit.

4. Module Deep-Dive (Continued)

4.3 Project Delivery & Visual Task Execution

- **Portfolio Management:** Manage project codes, budgets, target completion dates, team rosters, and client references.
- **Kanban State Machine:** Tasks progress through validated states: Todo → In Progress → In Review → Done. Enforces validations to prevent skipping stages.
- **Explicit Blocker Escalation:** Team members flag blockers directly on tasks (e.g. waiting for client assets). Blockers alert managers immediately to clear bottlenecks.
- **Rework & Cycle Time Metrics:** Monitors planned vs actual hours and rework loops. Frequent rework cycles surface requirements ambiguity or training needs.

4.4 Local Zero-Retention Multilingual Voice-to-Task Capture

- **Local Speech Transcription:** Powered by faster-whisper running on host infrastructure. Zero per-minute API costs.
- **Guaranteed Audio Shredding:** Ingested to ephemeral temp file, transcribed, and shredded immediately in a strict code finally block. Audio is never stored on disk or cloud.
- **11 Indian Regional Languages + English:** English, Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi, and Urdu.
- **Smart Multilingual NLP Parser:** Differentiates future tasks from completed work logs, converts spoken number words ("two hours", "दो घंटे") into numeric hours, and splits compound instructions ("Ayush ko frontend do aur Priya ko backend do").
- **Human Confirmation:** Generates an editable preview card for user verification before database commit.

4.5 Capability, Skills Matrix & Timed Assessments

- **Verified vs Self-Reported Skills:** Clear distinction between self-ratings (1-5) and peer/manager verified skills with proof.
- **Departmental Skill Heatmaps:** Visual matrix highlighting competency depth across teams and single-point dependency risks.
- **Role Benchmarks & Gap Analysis:** Compares employee skill profiles against designation requirements to guide training.
- **Timed Assessments:** Multiple-choice evaluations with randomized pools, time limits, and automated scoring.

4.6 Objective Performance Management & Snapshots

- **Cascading Goals & Measurable KPIs:** Quantifiable targets and weightings cascaded from company priorities down to individuals.
- **Structured 360 Reviews:** Multi-competency evaluation templates covering execution, communication, and leadership rubrics.
- **Deterministic Math Formula:** $\text{Final Score} = (\text{Goals} \times W_g) + (\text{KPIs} \times W_k) + (\text{Review Competencies} \times W_r)$. Non-negotiable grade bands: Exceeding ($\geq 90\%$), Meeting (75-89.9%), Needs Improvement (60-74.9%), Unsatisfactory ($< 60\%$).
- **Immutable Frozen Snapshots:** All inputs, ratings, and weights are locked into immutable cryptographic records preventing post-review alterations.

4. Module Deep-Dive: Commercial Operations & Security

4.7 Commercial Sales Intelligence & Geographic Management (GIS)

Connect sales execution directly to workforce capacity and geographic territories:

- **7-Tier Geographic Hierarchy (GIS):** Complete spatial hierarchy: Global → Country → State → District → City → Area → Pincode.
- **Interactive GIS Maps:** Visual heatmaps showing customer density, active leads, sales rep deployment, and revenue distribution.
- **Territory Quotas & Headcount Capacity:** Assign reps to territories with defined lead capacity thresholds to prevent lead neglect.
- **Lead Management & SLA Tracking:** Enforce response SLAs and automatically compute **Estimated Lost Revenue** for neglected leads.
- **Opportunity Pipeline & Deal Stages:** Visual pipeline (Prospecting to Closed-Won) with probability-weighted expected revenue.
- **Automated Revenue Ledger:** Closed-won opportunities automatically book realized revenue transactions onto the ledger.
- **3-Tier Commercial Access Scoping:** Reps see SELF records, managers see TEAM territories, and executives see ALL data.

4.8 Talent Acquisition & Automated Resume Screening

- **Job Description (JD) Library:** Standardized requisition templates tied to required organizational skill profiles.
- **Candidate Pipeline:** Visual stages from Sourced and Screened to Technical Interview, Offer, and Hired.
- **Automated Resume Screening:** Upload PDF/DOCX resumes. Extracts candidate skills, education, and experience years, matching them against job requirements with transparent match evidence for human recruiters.

4.9 Enterprise Governance, Security & Compliance

- **Fail-Closed Multi-Tenancy:** Logical tenant isolation enforced at database driver level via AsyncLocalStorage and Mongoose proxies. Zero cross-tenant data leakage (IDOR immunity).
- **HttpOnly Rotating Sessions:** 15-minute JWT access tokens paired with rotating refresh tokens. Replay detection revokes the token family.
- **Private Authenticated Document Storage:** Randomized keys in Cloudinary. Download links are signed, time-limited, and generated only after server-side permission checks. Files are never stored in public directories.
- **Tamper-Evident Audit Trail:** Every critical mutation (role changes, salary updates, voice commands, AI requests) records actor ID, timestamp, IP address, and JSON diffs.

4.10 Pragmatic & Ethical AI Assistance

- **Company Knowledge Assistant:** Answers employee policy questions grounded in uploaded company handbooks. Unhandled queries escalate to HR.
- **Advisory Review Drafting:** Prepares human-reviewable draft review summaries based on logged work history.
- **Multi-Provider Ready:** Pre-configured for Groq (llama-3.3-70b-versatile), with support for OpenAI, Anthropic, Gemini, or local models.

5. Feature Comparison: MobiusEMS vs. Disconnected SaaS Tools

Operational Domain	Fragmented SaaS Stack (Typical Setup)	MobiusEMS Unified Platform
Attendance Tracking	Biometric fingerprint machines; expensive hardware, fails for field staff, manual reconciliation.	Mobile browser Haversine geofencing (300m radius); zero hardware, anti-spoofing protection, automatic late/half-day logic.
Task Management	Standalone boards (Trello/Asana); disconnected from employee profiles and appraisals.	Integrated Kanban; directly feeds weekly deliverables, blocker logs, rework metrics, and review evidence.
Performance Reviews	Annual spreadsheets; subjective, recency-biased, lacking historical evidence.	Continuous evidence logging, measurable KPIs, 360 reviews, deterministic math scoring, and frozen snapshots.
Voice Task Capture	Manual web forms or expensive per-minute third-party cloud speech APIs.	Local faster-whisper on host; 11 Indian languages, zero per-minute fees, instant audio shredding.
Commercial Sales	Expensive standalone CRM; disconnected from employee capacity, leaves, and realized revenue.	Native 7-tier GIS hierarchy, territory quotas, lead SLA tracking, deal pipelines, and automated revenue ledger.
Recruitment & Screening	Manual resume scanning in shared inboxes or expensive external ATS subscriptions.	Integrated JD library, candidate pipelines, and automated PDF resume screening with advisory skill matching.
AI Capabilities	Black-box models or intrusive surveillance attempting to rate or rank staff automatically.	Transparent advisory AI; strict ethical lock against automated personnel decisions.

6. Implementation & 4-Week Rollout Plan

Week 1: Org Setup & Governance Provision isolated tenant workspace, configure departments, teams, designations, office coordinates, and assign administrative roles.	Week 2: Workforce & Daily Rhythm Bulk invite employees with resumable onboarding, activate geofenced attendance, and configure initial project Kanban boards.
Week 3: Sales & Skills Matrix Define geographic territories, assign sales reps, import customer accounts, and populate standardized skills catalog.	Week 4: Performance Cadence Configure KPI templates and cascading goals, launch weekly update cadence, and activate manager review workspaces.

7. Ready to Get Started?

Schedule a Tailored Demonstration: We will configure a live sandbox reflecting your department hierarchy, office geofences, and sales territories.

Contact: contact@whalexy.com | **Visit:** <https://employee.whalexy.com> | **Mobius Bloom Venture Pvt Ltd**